

Math 1019 — Definitions

A cumulative glossary, grouped by topic. Dates identify the lectures where the terms appear. Named laws and rules are included for reference. Wording and formulas have been clarified where the notes contain transcription errors.

1. Statements and propositions

September 11 and 14

Statement An assertion expressed in words or mathematical notation. Mathematical relations such as $=$ and $>$ can serve the role of a verb. Whether an assertion is a proposition depends on whether it has a definite truth value.

Declarative sentence A sentence that makes an assertion, such as “The grass is wet.” A declarative sentence is a proposition only if it is either true or false, but not both.

Imperative sentence A sentence that gives a command or instruction, such as “Hand in the assignment.” It is not a proposition.

Interrogative sentence A sentence that asks a question, such as “Are you in this class?” It is not a proposition.

Exclamatory sentence A sentence that expresses strong feeling or emphasis, such as “Wow!” An exclamation alone is not a proposition; an exclamatory sentence that also makes a definite assertion can be one.

Proposition A declarative statement that is either true or false, but not both. Its truth value need not be known to us. For example, $2 + 3 = 5$ is a true proposition, and $2 + 3 = 6$ is a false proposition.

Truth value The value true (T) or false (F) assigned to a proposition.

Propositional variable A symbol, usually p , q , or r , that represents a proposition. For example, p might represent “It is raining.”

Simple proposition A proposition treated as a single unit, without breaking it into propositions joined by logical connectives.

Compound proposition A proposition formed from other propositions using logical connectives, such as $\neg p$ or $p \wedge q$.

Logical connective An operation used to form a compound proposition. Its truth value is determined by the truth values of its inputs.

2. Logical connectives and truth tables

September 14 and 16

Negation (NOT) $\neg p$ means “not p .” It is true exactly when p is false.

Conjunction (AND) $p \wedge q$ means “ p and q .” It is true exactly when both p and q are true.

Disjunction (inclusive OR) $p \vee q$ means “ p or q .” It is true when at least one of p and q is true, including when both are true. This is the default meaning of OR in propositional logic.

Exclusive OR (XOR) $p \oplus q$ is true exactly when one of p and q is true and the other is false.

NAND The negation of a conjunction, $\neg(p \wedge q)$. It is false exactly when both inputs are true. (Named in the September 14 notes.)

Implication (conditional) $p \rightarrow q$ means “if p , then q .” It is false exactly when p is true and q is false; it is true in every other case. It rules out a true hypothesis with a false conclusion.

Hypothesis In $p \rightarrow q$, the proposition p . It is the condition assumed in the “if” part of the implication.

Conclusion In $p \rightarrow q$, the proposition q . It is the assertion in the “then” part. In an argument, the conclusion is the assertion that is meant to follow from the premises.

Biconditional (bi-implication) $p \leftrightarrow q$ means “ p if and only if q ,” abbreviated “ p iff q .” It is true exactly when p and q have the same truth value. Equivalently, both $p \rightarrow q$ and $q \rightarrow p$ hold.

Truth table A table listing every combination of truth values of the propositional variables and the resulting values of expressions built from them. With n distinct variables, there are 2^n rows of assignments.

Operator precedence The order in which connectives are applied when parentheses do not specify the grouping. The order given in the notes is $\neg$, then $\wedge$, then $\vee$, then $\rightarrow$, then $\leftrightarrow$.

For quick reference:

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

3. Related implications

September 16

Relative to the original implication $p \rightarrow q$:

Converse $q \rightarrow p$. Exchange the hypothesis and conclusion.

Inverse $\neg p \rightarrow \neg q$. Negate both the hypothesis and conclusion.

Contrapositive $\neg q \rightarrow \neg p$. Exchange and negate both parts. An implication is logically equivalent to its contrapositive.

The converse and inverse are logically equivalent to each other, but neither is generally equivalent to the original implication.

4. Classification and logical equivalence

September 16, 18, and 21

Tautology A propositional expression that is true for every assignment of truth values to its variables.

Example: $p \vee \neg p$.

Contradiction A propositional expression that is false for every assignment of truth values to its variables.

Example: $p \wedge \neg p$.

Contingency A propositional expression that is true for at least one assignment and false for at least one assignment. Example: $p \rightarrow q$.

Satisfiable expression An expression for which at least one assignment of truth values makes it true. Both tautologies and contingencies are satisfiable; contradictions are not.

Logical equivalence Expressions S and T are logically equivalent, written $S \equiv T$, if they have the same truth value under every assignment. Equivalently, $S \leftrightarrow T$ is a tautology. Equivalent subexpressions may be substituted for one another.

4.1. Named logical laws

September 18; extensions on September 21 and 23

Here T and F denote the constant truth values true and false.

Identity laws $p \wedge T \equiv p$; $p \vee F \equiv p$.

Domination laws $p \vee T \equiv T$; $p \wedge F \equiv F$.

Idempotent laws (idempotency) $p \wedge p \equiv p$; $p \vee p \equiv p$.

Double negation law $\neg(\neg p) \equiv p$.

Negation laws $p \vee \neg p \equiv T$; $p \wedge \neg p \equiv F$.

Commutative laws The order of the operands can be exchanged: $p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$.

Associative laws The grouping can be changed: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$; $(p \vee q) \vee r \equiv p \vee (q \vee r)$.

Distributive laws Each of AND and OR distributes over the other:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Absorption laws $p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$.

De Morgan's laws Negating a conjunction or disjunction negates each part and exchanges AND with OR:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

The same pattern applies to more than two propositions.

Conditional-disjunction equivalence (CD-equivalence) $(p \rightarrow q) \equiv (\neg p \vee q)$. Consequently, $\neg(p \rightarrow q) \equiv p \wedge \neg q$.

Contrapositive equivalence $(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$.

Biconditional equivalence $(p \leftrightarrow q) \equiv ((p \rightarrow q) \wedge (q \rightarrow p))$.

5. Predicates and quantifiers

September 21 and 23

Domain (of discourse) The collection of values a variable is allowed to take. Values must make the propositional function meaningful. For example, one possible domain for $x > 5$ is the integers.

Propositional function An expression such as $P(x)$ whose truth value depends on its variable. Substituting a particular value from the domain produces a proposition. For example, if $P(x)$ is $x + 7 = 13$, then $P(6)$ is true and $P(5)$ is false.

Subject The object or variable about which a property is asserted. In the September 21 example $Q(x)$, the subject is x .

Predicate The property asserted of the subject, such as “is greater than 5.” The notation $P(x)$ is also commonly called a predicate; predicates may involve more than one variable, as in $P(x, y)$.

Quantifier A symbol specifying how widely a predicate holds over a domain. The quantifiers introduced in the notes are $\forall$ and $\exists$.

Universal quantification $\forall xP(x)$ means “ $P(x)$ is true for every x in the domain.” It is false if there is even one domain element for which $P(x)$ is false.

Existential quantification $\exists xP(x)$ means “there is at least one x in the domain for which $P(x)$ is true.” It does not mean exactly one.

Counterexample An example that disproves a claim. For $\forall xP(x)$, it is a value a in the domain for which $P(a)$ is false.

Finite domain A domain with finitely many elements. For a domain consisting of $x_1, \dots, x_n$, quantification can be expanded as

$$\forall xP(x) \equiv P(x_1) \wedge \dots \wedge P(x_n)$$

$$\exists xP(x) \equiv P(x_1) \vee \dots \vee P(x_n)$$

Negation of quantifiers (De Morgan’s laws for quantifiers) Negating a quantified statement exchanges the quantifier and negates the predicate:

$$\neg(\forall xP(x)) \equiv \exists x(\neg P(x))$$

$$\neg(\exists xP(x)) \equiv \forall x(\neg P(x))$$

Nested quantifiers Multiple quantifiers applied to a predicate involving multiple variables. Read them from left to right. For example, $\forall x\exists yP(x, y)$ allows the choice of y to depend on x , whereas $\exists y\forall xP(x, y)$ requires one y that works for every x .

Commuting quantifiers Quantifiers commute if exchanging their order preserves the meaning. Two universal quantifiers commute, as do two existential quantifiers:

$$\forall x\forall yP(x, y) \equiv \forall y\forall xP(x, y)$$

$$\exists x\exists yP(x, y) \equiv \exists y\exists xP(x, y)$$

A universal and an existential quantifier generally cannot be exchanged.

Implicit universal quantifier A universal quantifier understood from context rather than written explicitly. For example, “If m and n are even integers, then $m + n$ is even” is a claim about all integers m and n . (September 25.)

5.1. Application from the notes: continuity

September 23

Continuity at a point A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at x_0 if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, (|x - x_0| < \delta \rightarrow |f(x) - f(x_0)| < \varepsilon).$$

Every positive error bound ε permits a positive distance bound δ that works for all x that close to x_0 . Its negation is

$$\exists \varepsilon > 0, \forall \delta > 0, \exists x \in \mathbb{R}, (|x - x_0| < \delta \wedge |f(x) - f(x_0)| \geq \varepsilon).$$

6. Arguments and rules of inference

September 23 and 25

Argument A collection of premises together with a proposed conclusion. The symbol $\therefore$ is read “therefore” and introduces the conclusion.

Premises (hypotheses) The propositions assumed or given in an argument, usually written $p_1, p_2, \dots, p_n$.

Valid argument An argument in which the premises cannot all be true while the conclusion is false. With premises $p_1, \dots, p_n$ and conclusion q , validity means

$$(p_1 \wedge \dots \wedge p_n) \rightarrow q$$

is a tautology. Validity concerns the inference; it does not assert that the premises are actually true.

Rule of inference A valid argument pattern that can be used to derive a conclusion from known or assumed premises.

Fallacy An invalid argument pattern. To show invalidity, give an assignment making every premise true and the conclusion false.

6.1. Rules used or named in the notes

Simplification From $p \wedge q$, infer p (or infer q).

Addition From p , infer $p \vee q$.

Conjunction rule From p and q as separate premises, infer $p \wedge q$.

Modus ponens From $p \rightarrow q$ and p , infer q .

Hypothetical syllogism From $p \rightarrow q$ and $q \rightarrow r$, infer $p \rightarrow r$.

Resolution From $p \vee q$ and $\neg p \vee r$, infer $q \vee r$. In the September 25 example, the remaining terms are both q , giving $q \vee q$, which is equivalent to q .

Universal instantiation From $\forall x P(x)$, infer $P(a)$ for any particular element a of the domain.

Universal modus ponens From $\forall x (P(x) \rightarrow Q(x))$ and $P(a)$, infer $Q(a)$, where a belongs to the domain.

Universal modus tollens From $\forall x (P(x) \rightarrow Q(x))$ and $\neg Q(a)$, infer $\neg P(a)$, where a belongs to the domain.

6.2. Common fallacies

Affirming the conclusion Incorrectly inferring p from $p \rightarrow q$ and q . Counterexample: p is false and q is true.

Denying the hypothesis Incorrectly inferring $\neg q$ from $p \rightarrow q$ and $\neg p$. Counterexample: p is false and q is true.

Affirming the disjunction Incorrectly inferring $\neg q$ from $p \vee q$ and p (or inferring $\neg p$ from $p \vee q$ and q). Counterexample: both p and q are true. Inclusive OR allows both.

7. Proof terminology and parity

September 25

Proof A valid mathematical argument establishing that a statement follows from the assumptions, definitions, and results being used.

Theorem A mathematical statement that has been proved, usually one regarded as important.

Proposition (as a proved result) A neutral name for a mathematical result that has been proved. This use differs from “proposition” in logic, where the statement may be true or false and need not be proved.

Lemma A proved result used as a stepping stone toward another result.

Corollary A proved result that follows readily from an established result.

Conjecture A statement proposed as true but not yet proved or disproved.

Even integer An integer n is even if and only if there exists an integer k such that $n = 2k$:

$$\exists k \in \mathbb{Z}, n = 2k.$$

Odd integer An integer n is odd if and only if there exists an integer k such that $n = 2k + 1$:

$$\exists k \in \mathbb{Z}, n = 2k + 1.$$

8. Corrections applied in this glossary

- September 16: the inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$. An implication excludes p true with q false.
- September 21: negating an existential quantifier gives $\forall x(\neg P(x))$. Existential quantification over a finite domain expands into a disjunction, rather than an arithmetic sum.
- September 23: $\neg(p \rightarrow q) \equiv p \wedge \neg q$. Negating a strict bound $|f(x) - f(x_0)| < \varepsilon$ gives $|f(x) - f(x_0)| \geq \varepsilon$.
- September 25: the usual “affirming the conclusion” fallacy includes both premises $p \rightarrow q$ and q .